

Supplementary Material

Digital Sovereignty under Pressure in Consumer Ecosystems (ICSOB 2026)

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This document provides the supplementary material accompanying the paper. Appendix A documents the review instruments (eligibility criteria and extraction framework). Appendix B presents the detailed profiles of the eleven literature-derived lock-in mechanisms, and Appendix C their coverage across the reviewed literature. Appendix D reproduces the complete survey instrument. Appendix E reports the descriptive statistics, Appendix F the exact agreement rates and 95 % Wilson confidence intervals underlying all validation decisions, and Appendix G the full validation matrix. The anonymized survey data and the Python analysis script that reproduces all reported figures are provided alongside this document.

A Review Instruments

The systematic literature review applied the inclusion and exclusion criteria shown in Table 1. The retained studies were analyzed through the four-field extraction framework shown in Table 2.

Table 1: Inclusion and exclusion criteria of the systematic literature review.

Criteria type	Description	Inclusion	Exclusion
Language	Articles are excluded based on language	English	Any other language
Length	Too short articles are excluded due to lack of necessary depth and detail	Length over 3 pages	Length under 3 pages
Type of literature	Articles are excluded if they fall into the category of grey literature	Peer-reviewed scientific literature and empirical or theoretical research papers	Reports, speeches, opinion papers
Academic relevance	Articles are excluded if the publication lacks sufficient academic reputation or visibility	Peer-reviewed or highly cited sources	Not peer-reviewed or low citation count
Thematic relevance	Articles are excluded if they do not address the core aspects of the research focus	Studies addressing specific lock-in / switching barriers in consumer digital ecosystems	Enterprise-only focus, state-level only, or insufficient detail on lock-in mechanisms

Table 2: Extraction template

Field	Content
Mechanism name	Short, standardized label for the mechanism, harmonized across papers describing the same mechanism.
Explanation	Faithful paraphrase of how the mechanism is described across the contributing papers.
Effect	Description of how the mechanism generates switching costs, articulated in terms of the procedural, financial, and relational dimensions of Burnham et al. (2003) where applicable.
Source(s)	Bibliographic reference(s) of the study or studies in which the mechanism is identified.

B Detailed Lock-in Mechanism Profiles

This appendix presents the detailed profile of each of the eleven lock-in mechanisms retrieved from the systematic literature review. Each profile describes the mechanism, its effect on switching costs, and the literature sources from which it was extracted.

M1: Seamless coordination between products (*Experiential*)

Through software and data synchronization, the ecosystem connects its native products into a cohesive system in which all components inherently belong and work together as one organic whole. Each new product the user adds introduces further ties, so the network grows. The experiential coordination value of any device, therefore, depends on its place in the ecosystem rather than on its standalone properties.

Effect: A product that is switched to an external ecosystem falls into a structural hole because it is severed from the seamless coordination. The user loses the ties associated with that connection, and the remaining devices lose a portion of the coordination value they had drawn from the now-smaller network. Therefore, switching imposes procedural costs through the time and effort needed to identify, learn, and configure substitute synchronization paths. Financial costs through the loss of value accumulated within the coordinated network and relational costs, as the experience of one's products inherently belonging together is broken by the introduction of an outside device.

Sources: (Chang, 2025; Crawford, 2024; Kim and Shin, 2017; Ramadan, 2021)

M2: Cross-device interface and experience continuity (*Experiential*)

The ecosystem orchestrator applies a continuity in design terms across the ecosystem. This happens through a shared visual idiom, consistent interaction conventions, and a uniform design language. Multiple products thereby behave as a single coherent environment in which the user encounters the same interaction logic regardless of which product is used.

Effect: Through the common experience, each product within the ecosystem deepens the user's familiarity and strengthens the routines that depend on it until they become automatic. When one product is replaced by an alternative outside the ecosystem, the user must abandon accumulated patterns and relearn an unfamiliar interaction language while still operating legacy devices from the previous ecosystem. Therefore, switching imposes procedural costs through the time and effort required to relearn interface conventions. Financial costs through the forfeiture of the learning investment accumulated in the previous ecosystem's interface, and relational costs through detachment from the platform owner's design language with which the user has come to identify.

Sources: (Crawford, 2024; Polites and Karahanna, 2012; Ramadan, 2021)

M3: Ecosystem-exclusive first-party products (*Product-level*)

Selected first-party products operate exclusively within the orchestrators operating systems and cannot be installed, accessed, or carried over to competing ones.

Effect: Because some first-party products cannot be carried over or be accessed on a competing ecosystem, the user's ability to maintain these services outside the ecosystem is foreclosed. The user therefore faces procedural costs for the time and effort to identify, learn, and configure equivalent services on the new platform. Financial costs through the loss of the products themselves, and relational costs through detachment from the ecosystem-specific service experiences.

Sources: (Crawford, 2024; Wörsdörfer, 2024; Montgomerie and Roscoe, 2013; Uyar et al., 2025)

M4: Interdependent first-party products (*Product-level*)

Selected products are designed to depend on other products from the same ecosystem to function. If the base product on which they depend is replaced by a non-ecosystem alternative, the dependent products lose their function or are reduced to substantially limited operation, because the ecosystem orchestrator has technically restricted their compatibility to the ecosystem's own components.

Effect: When a user who owns interdependent products switches the base product, the dependent products lose their value or operate only at substantially reduced functionality. Consequently, switching incurs additional financial costs from the loss of value in unusable dependent products, and additional procedural costs associated with the time and effort needed to identify and configure substitute products that are compatible with the new ecosystem.

Sources: (Wörsdörfer, 2024; Haftor et al., 2025; Chang, 2025)

M5: Multi-product subscription bundle (*Commercial*)

The ecosystem owner offers a subscription bundle that pairs multiple products at a price below the sum

of their individual rates. The bundle is offered in scaled tiers that support either single-user or shared household use, while all tiers grant private access to each product for all members. The bundle typically combines products that would otherwise have functional substitutes available on competing platforms with products exclusive to the ecosystem orchestrator.

Effect: The bundle aims to move the user more fully into the ecosystem. If a user switches to another ecosystem, the change affects multiple bundled products at the same time and, where the bundle is held jointly with other household members, multiple persons. In addition, the first-party products that are exclusive to the ecosystem cannot be carried over and must be replaced through independent providers. Therefore, switching imposes procedural costs through the time and effort needed to identify, learn, and configure replacements for each bundled product in turn. Financial costs from the loss of the bundled discount, and relational costs from the coordination required with co-subscribers in household tiers, where switching requires the agreement and adjustment of all bundle members, and from the loss of the brand-curated experience the bundle had assembled.

Sources: (Crawford, 2024; Wörsdörfer, 2024)

M6: Cross-promotional trials at product purchase (*Commercial*)

A discounting practice in which the purchase of an ecosystem product automatically grants the user a free trial of one or more additional ecosystem products. The ecosystem orchestrator foregoes subscription fees in the short term in order to introduce users to additional products at the moment of acquisition.

Effect: The trial aims to introduce the user to additional products and move them more fully into the ecosystem. Over the free trial period, the user becomes familiar with the trial products and incorporates them into daily routines. After the trial expires, the user typically continues to subscribe at the regular rate, now habituated to the products and embedded more deeply within the ecosystem than at the moment of acquisition. Switching to an external alternative, therefore, imposes higher procedural costs because of the additional products now in use, the time and effort required to identify, learn, and configure equivalents is higher. Furthermore, switching imposes relational costs by detaching the user from the brand-curated product experiences with which the user has come to identify during the trial and post-trial period.

Sources: (Crawford, 2024; Wörsdörfer, 2024)

M7: Limiting third-party access to ecosystem products (*Marketplace*)

Specific features of the products in the ecosystem are reserved for the ecosystem orchestrator’s first-party services and are not available to third-party developers. As a result, the user has no functional third-party alternative to the first-party services for these features within the ecosystem.

Effect: Users can interact only with the ecosystem orchestrator’s first-party features and accumulate credentials, settings, and consumption routines exclusively within them. When switching to a competing ecosystem, these arrangements cannot be migrated and must be reconstructed from scratch in third-party services, with no prior multihoming experience to draw on. Switching, therefore, imposes procedural costs through the time and effort required to identify, learn, and configure equivalent third-party services on the new ecosystem. Financial costs through the loss of the credentials, settings, and consumption routines that the user has accumulated against the first-party services and that cannot be carried over. Relational costs through the loss of the familiarity with the first-party services on which the user has come to rely.

Sources: (Wörsdörfer, 2024; Todd, 2018)

M8: Preferencing of first-party products (*Marketplace*)

First-party products are promoted by the ecosystem orchestrator in its own stores above competing third-party alternatives. Furthermore, first-party products are pre-installed, and several of them cannot be uninstalled. Additionally, the default settings automatically route user actions to first-party products.

Effect: Through the combination of in-store ranking, system-level defaults, and the privileged integration that first-party products receive within the ecosystem, the user adopts more first-party products than they would have chosen on their own and thus embeds themselves more deeply within the ecosystem. Switching to a competing ecosystem therefore imposes amplified procedural costs through the time and effort needed to identify, learn, and configure third-party equivalents for the larger first-party set the user has accumulated, and amplified relational costs through the broader brand-identification developed through sustained default-driven use.

Sources: (Wörsdörfer, 2024; Todd, 2018; Uyar et al., 2025)

M9: Centralized ecosystem-exclusive account (*Identity & Asset*)

A single account is required to access the ecosystem’s products, and it is not transferable to other ecosystems. It consolidates various types of sensitive user information under one identifier.

Effect: Switching to a competing ecosystem requires the user to abandon the account and set up a new one from scratch in the new ecosystem. Switching therefore imposes procedural costs through the time and effort needed to re-establish an equivalent account on the new ecosystem. Financial costs through the loss of the data and content accumulated against the account and relational costs through the loss of the digital identity cultivated around the account.

Sources: (Crawford, 2024; Krämer, 2026; Rong et al., 2026)

M10: Non-portable purchased digital content (*Identity & Asset*)

Digital content purchased within the ecosystem is bound to it through proprietary formats or digital rights management, and cannot be transferred to or used on competing ecosystems.

Effect: Switching to a competing ecosystem leaves the accumulated content library tied to the original platform, since the purchased content cannot be carried over. Switching, therefore, imposes financial costs through the loss of the digital purchases accumulated over time, and procedural costs through the time and effort needed to re-establish an equivalent library on the new platform.

Sources: (Jacobides and Lianos, 2021; Wörsdörfer, 2024; Montgomerie and Roscoe, 2013)

M11: Personified AI assistant (*Identity & Asset*)

The ecosystem provides a humanized AI assistant as a central interaction channel with its products. Through repeated human-like exchanges, the assistant adapts to the user’s voice, preferences, and conversational history, fostering affective bonding.

Effect: Over time, the user builds an affective bond with the assistant and, through it, with the ecosystem the assistant mediates. Switching to a competing ecosystem, therefore, imposes procedural costs through the time and effort needed to acclimate to a different assistant and recreate the personalized settings accumulated against the previous one, and relational costs through detachment from the affective bond and the self-identification cultivated around the personified assistant.

Sources: (Ramadan, 2021; Alimamy and Kuhail, 2023)

C Coverage of the Mechanisms across the Reviewed Literature

Table 3 reports which of the fourteen reviewed studies describe which mechanism.

Table 3: Coverage of the lock-in mechanisms across the reviewed literature.

Mechanism	Reviewed articles													
	(Alimamy and Kuhail, 2023)	(Chang, 2025)	(Crawford, 2024)	(Haftor et al., 2025)	(Jacobides and Lianos, 2021)	(Kim and Shin, 2017)	(Krämer, 2026)	(Montgomerie and Roscoe, 2013)	(Polites and Karahanna, 2012)	(Ramadan, 2021)	(Rong et al., 2026)	(Todd, 2018)	(Uyar et al., 2025)	(Wörsdörfer, 2024)
M1 Seamless coordination between products		×	×			×				×				
M2 Cross-device interface and experience continuity			×						×	×				
M3 Ecosystem-exclusive first-party products			×					×					×	×
M4 Interdependent first-party products		×		×										×
M5 Multi-product subscription bundle			×											×
M6 Cross-promotional trials at product purchase			×											×
M7 Limiting third-party access to ecosystem products												×		×
M8 Preferencing of first-party products												×	×	×
M9 Centralized ecosystem-exclusive account			×				×				×			
M10 Non-portable purchased digital content					×			×						×
M11 Personified AI assistant	×									×				
Mechanisms mentioned	1	2	6	1	1	1	1	2	1	3	1	2	2	7

D Survey Instrument

This appendix presents the complete survey instrument used for the empirical validation of the lock-in mechanisms identified in Section 4 of the paper. The survey was administered online and structured as a sequence of screens. Screen 1 contains the welcome text, informed consent, ecosystem definition, and ecosystem selection. Screens 2–12 each present one of the eleven lock-in mechanisms (M1–M11) through a paired-item structure: a Likert item eliciting the respondent’s agreement that the mechanism acts as a switching barrier, followed by a multi-select item identifying which of the procedural, financial, or relational cost types weighs on the respondent. The follow-up item was shown to all respondents but analytically retained only for those who agreed with the Likert item (response 4 or 5). The option “This is not a feature of the ecosystem I use” records non-applicability, which is treated separately from disagreement.

Screen 1: Introduction and Ecosystem Definition

This survey is part of a master’s thesis at the University of Amsterdam. It takes approximately seven minutes to complete and is fully anonymous, there is no personal information stored. By continuing, you confirm that you are at least 18 years old and consent to the use of your responses for academic research.

This survey is about consumer digital ecosystems and their end consumers. The term ecosystem refers to an integrated set of hardware devices, software, and services from a single company, organized around one or more central platforms and complemented by third-party apps and services. The most common examples are:

<i>Apple</i>	iPhone, Mac, iPad, Apple Watch, iCloud, Apple Music
<i>Google / Android</i>	Pixel or other Android phones, Chromebook, Google Drive
<i>Amazon</i>	Echo, Fire devices, Kindle, Prime Video, Amazon Music
<i>Microsoft</i>	Windows PC, Surface, Xbox, OneDrive
<i>Samsung</i>	Galaxy phones, Galaxy Watch, Galaxy Buds, Samsung TV
Question	Which of these ecosystems do you use in your personal digital life?
Response options	☐ Apple ☐ Google / Android ☐ Amazon ☐ Microsoft ☐ Samsung ☐ Other ☐ I do not consider myself part of any ecosystem
Skip logic	<i>If “I do not consider myself part of any ecosystem” is selected, the survey ends with a thank-you message.</i>

Table 4: Ecosystem examples and item S1.

Screen 2: Mechanism 1 — Seamless coordination between products

F1.1	Likert item
Question	The products in my ecosystem (such as my phone, computer or apps) automatically share data and continue tasks across each other. For example, copying text from one to another, picking up a call on a different device, or accessing the same files on all of them. Because a product from a different ecosystem would not work together with my other products this way, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F1.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to set up similar connections between new products ☐ Losing the benefits I now get from having my products work together ☐ The loss of the feeling that my products belong together as one coherent set

Table 5: Survey items M1.1 and M1.2 — Seamless coordination between products.

Screen 3: Mechanism 2 — Cross-device interface and experience continuity

F2.1	Likert item
Question	The products of my ecosystem (such as my phone, computer, or apps) share a similar look-and-feel and similar ways of doing things. For example, the same gestures, comparable menus, and consistent settings layouts. Because a product from a different ecosystem would have different conventions and require me to relearn how to use it, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F2.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to relearn how a different ecosystem’s products work ☐ The loss of the learning effort I have already invested in my current ecosystem ☐ The loss of the familiar style of my current ecosystem that I have come to identify with

Table 6: Survey items M2.1 and M2.2 — Cross-device interface and experience continuity.

Screen 4: Mechanism 3 — Ecosystem-exclusive first-party products

F3.1	Likert item
Question	Some of the apps or services I use in my ecosystem are available only within it. They cannot be installed on products from other ecosystems, or they only work fully on my current brand's products. For example, certain apps, payment services, or cloud storage that come built into my products. Because these apps and services would not be available if I switched to a different ecosystem, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F3.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to find and learn alternative apps or services on a different ecosystem ☐ The loss of these specific apps and services that I currently use ☐ The loss of the distinctive experience of these apps and services that I have come to identify with

Table 7: Survey items M3.1 and M3.2 — Ecosystem-exclusive first-party products.

Screen 5: Mechanism 4 — Interdependent first-party products

F4.1	Likert item
Question	Some of my products are designed to work fully only when combined with other products from this ecosystem. For example, a smartwatch that needs my phone to work fully, wireless earbuds whose features are limited on other brands, or a pen that works only with its matching tablet. Because these dependent products would lose their full function if I replaced the base product with one from a different ecosystem, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F4.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to find replacement products compatible with a different ecosystem ☐ The loss of value in the dependent products that would no longer work properly

Table 8: Survey items M4.1 and M4.2 — Interdependent first-party products.

Screen 6: Mechanism 5 — Multi-product subscription bundle

F5.1	Likert item
Question	My ecosystem offers a subscription bundle that combines multiple services (such as cloud storage, music streaming, video streaming, gaming, or fitness) into a single plan at a lower price than buying each separately. Some bundles can also be shared with family or household members. Because switching to a different ecosystem would require replacing each bundled service separately and at full price, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F5.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to find and set up separate replacements for each bundled service ☐ The loss of the bundled discount, since buying services separately would cost more ☐ The coordination needed with family or household members who share my bundle

Table 9: Survey items M5.1 and M5.2 — Multi-product subscription bundle.

Screen 7: Mechanism 6 — Cross-promotional trials at product purchase

F6.1	Likert item
Question	When I bought products of the ecosystem (such as my phone, watch, or speaker), I received free trials of additional services such as music streaming, video streaming, fitness, or gaming. I integrated these services into my daily routines during the trial period and continued subscribing afterward. Because switching to a different ecosystem would now mean replacing not just my products but also these additional services I have come to use, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F6.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to find and set up replacements for the additional services I now use ☐ The loss of the familiar experience of the additional services I tried and have come to identify with

Table 10: Survey items M6.1 and M6.2 — Cross-promotional trials at product purchase.

Screen 8: Mechanism 7 — Limiting third-party access to ecosystem products

F7.1	Likert item
Question	Some features of my products in this ecosystem (such as contactless payments, voice assistant integration, or specific hardware controls) are reserved for the brand's own apps and cannot be used by third-party apps. As a result, I rely on the brand's own services for these functions, and there is no functional alternative within this ecosystem. Because switching to a different ecosystem would mean rebuilding these arrangements with third-party services I have not used before, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F7.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to find and learn third-party alternatives within a different ecosystem ☐ The loss of credentials, settings, or routines I have built up in the ecosystem's first-party services ☐ The loss of the familiarity with the first-party services I have come to rely on

Table 11: Survey items M7.1 and M7.2 — Limiting third-party access to ecosystem products.

Screen 9: Mechanism 8 — Preferencing of first-party products

F8.1	Likert item
Question	In the ecosystem I use, the brand’s own apps and services are preinstalled on my products (some cannot be uninstalled), set as the defaults for common actions (such as the default browser, maps, or music player), and shown at the top when I search in the brand’s store. As a result, I use more of the brand’s own apps and services than I might have chosen on my own. Because switching to a different ecosystem would mean replacing this larger set of apps and services I have come to rely on, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F8.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to find and set up third-party alternatives for the brand’s apps I now use ☐ The loss of the familiar use of the brand’s apps that I have come to identify with through default-driven use

Table 12: Survey items M8.1 and M8.2 — Preferencing of first-party products.

Screen 10: Mechanism 9 — Centralized ecosystem-exclusive account

F9.1	Likert item
Question	My ecosystem requires a single account (such as an Apple ID, Google Account, Amazon Account, or Microsoft Account) that connects all my products. This account stores my personal information, settings, app history, photos, and other data I have accumulated over time. Because this account cannot be transferred to a different ecosystem and the accumulated data would not move with me, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F9.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to set up a new account on a different ecosystem ☐ The loss of the data, settings, and content I have accumulated against my current account ☐ The loss of the digital identity I have built up around my account over time

Table 13: Survey items M9.1 and M9.2 — Centralized ecosystem-exclusive account.

Screen 11: Mechanism 10 — Non-portable purchased digital content

F10.1	Likert item
Question	Digital content I have bought through this ecosystem (such as ebooks, movies, music, or games) is bound to it through the platform’s own format or digital rights protection. This content cannot be transferred to or used on a different ecosystem. Because the digital purchases I have built up over time would not move with me to a different ecosystem, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F10.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to rebuild an equivalent content library on a different ecosystem ☐ The loss of the digital content I have purchased and accumulated over time

Table 14: Survey items M10.1 and M10.2 — Non-portable purchased digital content.

Screen 12: Mechanism 11 — Personified AI assistant

F11.1	Likert item
Question	The ecosystem I use has its own AI assistant (such as Alexa, Siri, Google Assistant) with a distinct name, voice, and personality that I interact with regularly. Over time, through repeated human-like exchanges, the assistant has adapted to my voice, preferences, and routines, and I have come to find its way of interacting familiar. Because a different ecosystem’s assistant would have a different name, voice, and personality, and would not know my preferences and routines, switching some of my products would be difficult for me.
Response scale	1 = Strongly disagree 2 = Disagree 3 = Neither agree nor disagree 4 = Agree 5 = Strongly agree
Or	☐ This is not a feature of the ecosystem I use
F11.2	Follow-up (multi-select)
Question	What specifically would weigh on you in the case of switching? Select all that apply.
Response options	☐ The time and effort to acclimate to a different assistant and recreate my personalized settings ☐ The loss of the familiar way of interacting with the assistant that I have come to identify with

Table 15: Survey items M11.1 and M11.2 — Personified AI assistant.

E Descriptive Statistics

Table 16 reports the per-mechanism descriptive statistics of the existence item.

Table 16: Descriptive statistics per lock-in mechanism. n_{app} : valid Likert responses; n_{NA} : respondents marking the mechanism as not a feature of their ecosystem; n_{agr} : respondents who agreed or strongly agreed (Likert 4 or 5); Rate: n_{agr}/n_{app} .

Mechanism	n_{app}	n_{NA}	M	SD	Med	n_{agr}	Rate
M1 Seamless coordination between products	217	5	3.88	1.03	4	171	78.8 %
M2 Cross-device interface and experience continuity	217	5	3.90	1.00	4	173	79.7 %
M3 Ecosystem-exclusive first-party products	218	4	3.83	1.06	4	159	72.9 %
M4 Interdependent first-party products	210	12	3.73	1.10	4	144	68.6 %
M5 Multi-product subscription bundle	199	23	3.58	1.16	4	131	65.8 %
M6 Cross-promotional trials at product purchase	201	21	3.32	1.18	4	113	56.2 %
M7 Limiting third-party access to ecosystem products	208	14	3.78	1.01	4	148	71.2 %
M8 Preferencing of first-party products	219	3	3.67	1.16	4	155	70.8 %
M9 Centralized ecosystem-exclusive account	217	5	4.18	0.97	4	182	83.9 %
M10 Non-portable purchased digital content	202	20	3.39	1.19	4	114	56.4 %
M11 Personified AI assistant	195	27	3.21	1.35	3	97	49.7 %

F Wilson Confidence Intervals

This appendix reports the exact figures behind the validation decisions presented in Section 4.2 of the paper.

Table 17: Exact 95 % Wilson score confidence intervals and validation decisions per lock-in mechanism.

Mechanism	Rate	95 % Wilson CI	Decision
M1 Seamless coordination	78.8	[72.9, 83.7]	Robust
M2 Cross-device continuity	79.7	[73.9, 84.5]	Robust
M3 Ecosystem-exclusive 1P	72.9	[66.7, 78.4]	Robust
M4 Interdependent 1P	68.6	[62.0, 74.5]	Robust
M5 Multi-product bundle	65.8	[59.0, 72.1]	Robust
M6 Cross-promotional trials	56.2	[49.3, 62.9]	Marginal
M7 Limiting third-party access	71.2	[64.7, 76.9]	Robust
M8 Preferencing first-party	70.8	[64.4, 76.4]	Robust
M9 Centralized account	83.9	[78.4, 88.2]	Robust
M10 Non-portable content	56.4	[49.5, 63.1]	Marginal
M11 Personified AI assistant	49.7	[42.8, 56.7]	Not validated

Table 18: Exact 95 % Wilson score confidence intervals and validation decisions per Burnham et al. (2003) dimension.

Mechanism	Dimension	Rate	95 % Wilson CI	Decision
M1 Seamless coordination	Procedural	70.0	[62.7, 76.4]	Robust
	Financial	71.2	[64.0, 77.5]	Robust
	Relational	45.3	[38.0, 52.8]	Not corroborated
M2 Cross-device continuity	Procedural	69.9	[62.7, 76.3]	Robust
	Financial	45.7	[38.4, 53.1]	Not corroborated
	Relational	65.9	[58.6, 72.5]	Robust
M3 Ecosystem-exclusive 1P	Procedural	57.2	[49.5, 64.7]	Marginal
	Financial	71.7	[64.2, 78.1]	Robust
	Relational	49.1	[41.4, 56.8]	Not corroborated
M4 Interdependent 1P	Procedural	62.2	[54.1, 69.8]	Robust
	Financial	66.4	[58.4, 73.7]	Robust
M5 Multi-product bundle	Procedural	55.0	[46.4, 63.2]	Marginal
	Financial	67.2	[58.7, 74.6]	Robust
	Relational	48.1	[39.7, 56.6]	Not corroborated
M6 Cross-promotional trials	Procedural	57.7	[48.4, 66.4]	Marginal
	Relational	73.0	[64.0, 80.4]	Robust
M7 Limiting third-party access	Procedural	61.2	[53.2, 68.7]	Robust
	Financial	70.7	[62.9, 77.5]	Robust
	Relational	60.5	[52.5, 68.1]	Robust
M8 Preferencing first-party	Procedural	65.6	[57.7, 72.7]	Robust
	Relational	72.2	[64.6, 78.7]	Robust
M9 Centralized account	Procedural	63.7	[56.4, 70.4]	Robust
	Financial	86.0	[80.2, 90.4]	Robust
	Relational	55.9	[48.5, 62.9]	Marginal
M10 Non-portable content	Procedural	64.6	[55.4, 72.8]	Robust
	Financial	79.6	[71.3, 86.0]	Robust

G Full Validation Matrix

For methodological transparency, Table 19 reports the full per-mechanism, per-dimension validation matrix underlying the Lock-in Catalog (Table 1 of the paper). Symbols indicate the Wilson-interval classification: ✓ robust (lower bound above 50 %), (✓) marginal (interval straddles 50 %), × not corroborated (sample rate below 50 %), and dimension not attributed in the catalog. For M11 the H_2 columns are not evaluated since H_1 is not validated.

Table 19: Joint validation outcome per lock-in mechanism. ✓: robust; (✓): marginal; ×: not corroborated.

Mechanism	H ₁	H ₂		
		Procedural	Financial	Relational
<i>Experiential</i>				
M1 Seamless coordination between products	✓	✓	✓	×
M2 Cross-device interface and experience continuity	✓	✓	×	✓
<i>Product-level</i>				
M3 Ecosystem-exclusive first-party products	✓	(✓)	✓	×
M4 Interdependent first-party products	✓	✓	✓	—
<i>Commercial</i>				
M5 Multi-product subscription bundle	✓	(✓)	✓	×
M6 Cross-promotional trials at product purchase	(✓)	(✓)	—	✓
<i>Marketplace</i>				
M7 Limiting third-party access to ecosystem products	✓	✓	✓	✓
M8 Preferencing of first-party products	✓	✓	—	✓
<i>Identity & asset</i>				
M9 Centralized ecosystem-exclusive account	✓	✓	✓	(✓)
M10 Non-portable purchased digital content	(✓)	✓	✓	—
M11 Personified AI assistant	×	—	—	—

References

- Alimamy, S. and Kuhail, M. A. (2023). I will be with you Alexa! the impact of intelligent virtual assistant’s authenticity and personalization on user reuse intentions. *Computers in Human Behavior*, 143.
- Burnham, T. A., Frels, J. K., and Mahajan, V. (2003). Consumer switching costs: A typology, antecedents, and consequences. *Journal of the Academy of Marketing Science*, 31(2):109–126.
- Chang, J. H. (2025). Secret power of the product ecosystem: A network perspective from the case of Apple. *Journal of Business Research*, 200.
- Crawford, C. (2024). TV plus what? data-capital, brand extension, and lock-in effects in the Apple ecosystem. *New Review of Film and Television Studies*, 22(4):1061–1082.
- Haftor, D. M., Climent, R. C., and Kallmuenzer, A. (2025). Business ecosystems as a way to activate lock-in in business models: a theoretical integration. *International Entrepreneurship and Management Journal*, 21(1).
- Jacobides, M. G. and Lianos, I. (2021). Ecosystems and competition law in theory and practice. *Industrial and Corporate Change*, 30(5):1199–1229.
- Kim, H. and Shin, D. H. (2017). The effects of platform as a technology standard on platform-based repurchases. *Digital Policy, Regulation and Governance*, 19(2):41–55.

- Krämer, J. (2026). Balancing privacy and platform power in the mobile ecosystem: The case of Apple’s App Tracking Transparency. *Computer Law & Security Review*, 60:106255.
- Montgomerie, J. and Roscoe, S. (2013). Owning the consumer—getting to the core of the Apple business model. *Accounting Forum*, 37(4):290–299.
- Polites, G. and Karahanna, E. (2012). Shackled to the status quo: The inhibiting effects of incumbent system habit, switching costs, and inertia on new system acceptance. *MIS Quarterly*, 36:21–42.
- Ramadan, Z. B. (2021). “Alexafying” shoppers: The examination of Amazon’s captive relationship strategy. *Journal of Retailing and Consumer Services*, 62.
- Rong, K., Zhao, Y., Kang, Z., and Parente, R. (2026). All-in-one strategy of platforms: Crafting multi-product ecosystems. *Technovation*, 150.
- Todd, P. F. (2018). Intra-platform exclusion in software markets. *Journal of Antitrust Enforcement*, 6(3):409–436.
- Uyar, L., Clement, M., Veseli, B., and Kandziora, M. (2025). The rise of smart speakers: a comparative analysis of music streaming service adoption and platform interdependencies. *Journal of Media Economics*, 37(3):129–147.
- Wörsdörfer, M. (2024). Apple’s antitrust paradox. *European Competition Journal*, 20(1):113–146.